

Project Idea - Group 8

Title : E-Learning Platform for Self-Learning Using Large Language Models (LearnMateAI)

Introduction :

The proposed system is an e-learning platform that transforms uploaded lecture materials into an interactive self-learning experience. Students can upload course materials in PDF format, after which the system automatically extracts the content and uses a multi-agent Large Language Model (LLM) architecture to generate educational resources that support independent learning. The primary objective is to facilitate independent study by automatically providing supplementary materials like summaries, quizzes, and contextual question-answering chat functions based specifically on the user's uploaded documents.

Proposed Solution :

The solution will be a full-stack web application featuring a React frontend, a Python FastAPI backend, MongoDB NoSQL database and Vector Database. The core functionality is driven by a PDF processing pipeline together with LLM usage:

- **PDF Processing Pipeline:** Uploaded standard lecture materials undergo validation, metadata extraction, and layout-aware parsing to accurately capture text, images, and tables. The document is then cleaned, structurally reconstructed, and divided using intelligent semantic chunking.
- **Multi-Agent LLM Architecture:** The platform utilizes three distinct LLM roles for optimized performance and cost:
 - a. **Embedding Generation:** To convert semantic chunks into high-dimensional vectors stored in a vector database. This allows for efficient similarity searches without reading the entire PDF every time.
 - b. **Resource Generation:** High-reasoning models (e.g., Gemini 1.5 Pro or GPT-4o) process the chunks to generate lecture summaries, key points, flashcards, MCQs, and practice questions.
 - c. **RAG Chat Engine:** Fast, low-latency models (e.g., Gemini 1.5 Flash) power an AI chatbot that answers student questions by retrieving the top-K relevant chunks from the vector database to construct contextual prompts.
- **User Features:** The platform will include user authentication, Simple and user friendly user dashboard for PDF management, dedicated sections to access generated resources, progress tracking and to interact with question-answering chat applications.
- **Content Intelligence layer - Training ML models to classify by each topic.**

Datasets:

Standard educational materials and lecture notes in PDF format. PDF can contain text, images, tables and graphs. (Initially, the system is not targeting question papers, tutorials, or research papers)

Similar Projects:

1. <https://github.com/Tejas-00/StudyGenie-AI-Powered-Personal-Tutor>
2. <https://github.com/devcom33/Chat-with-Your-Documents>
3. <https://github.com/BhaveshBhakta/PDFQuizzer-A-Smart-PDF-MCQ-Generator>
4. <https://github.com/Adityathere/ChatPDF>
5. <https://www.quiz-genie.com/> - <https://github.com/SudarshanaSRao/QuizGenie>
6. <https://github.com/gkuling/QuizGen-RAG>

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